

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).

Assessment Tool Weightage: 25%

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QUESTIONS: 10

Total Marks: 50

Annexure A

Data Interpretation

Duration: 1hr

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

a. Introduction

In Reinforcement Learning (RL), an agent learns to make better decisions by interacting with an environment. It receives rewards based on the quality of its actions and updates its policy through repeated experience. The reward obtained over different episodes is an important measure of the agent's learning performance.

b. Observed Reward Trend

In the given scenario, the agent's reward gradually increases from **10 to 45** over multiple training episodes. This positive trend indicates that the agent is improving its decisions as training progresses. Initially, the agent has limited knowledge and may select ineffective actions, resulting in lower rewards.

c. Learning Performance

As the number of episodes increases, the agent gains more experience about the environment. It learns which actions produce positive outcomes and which actions should be avoided. The increase in reward from **10 to 45** therefore indicates improved learning performance. The agent is becoming better at maximizing its cumulative reward.

d. Policy Improvement

The increasing reward suggests that the agent's **policy is improving**. A policy determines the action selected by the agent for a given state. Through repeated interaction, the agent updates its policy to prefer actions that provide higher rewards. Thus, the higher reward indicates that the learned policy is becoming more effective.

e. Convergence

If the reward eventually becomes stable around a high value, it may indicate that the agent's policy has reached **convergence**. However, an increase from 10 to 45 alone does not prove that the policy is optimal. The agent could have learned a good but suboptimal policy.

f. Conclusion

Overall, the gradual increase in reward from **10 to 45** demonstrates successful learning, improved decision-making, and policy refinement. Further evaluation using average reward, success rate, and comparison with the maximum possible reward would be required to confirm optimality.

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

a. Introduction

In Reinforcement Learning, the **exploration–exploitation trade-off** determines how an agent balances trying new actions with selecting actions that are already known to provide good rewards. The ϵ -greedy strategy uses the parameter ϵ (**epsilon**) to control this balance.

b. Effect of Epsilon

In the given scenario, the agent is tested with $\epsilon = 0.1, 0.3, \text{ and } 0.5$. A higher epsilon means more exploration, while a lower epsilon means greater exploitation. With $\epsilon = 0.1$, the agent explores approximately 10% of the time and exploits the best-known action about 90% of the time. With $\epsilon = 0.5$, exploration increases to approximately 50%.

c. Reward Comparison

The cumulative reward is highest when $\epsilon = 0.1$ and lowest when $\epsilon = 0.5$. This indicates that the environment rewards consistent exploitation of already learned actions. Excessive exploration causes the agent to select uncertain or less rewarding actions more frequently, reducing its cumulative reward.

d. Learning Interpretation

The result suggests that the agent has already learned useful actions and therefore benefits from exploiting its current knowledge. However, exploration is still important during early training because it allows the agent to discover potentially better actions. A very high epsilon may prevent the agent from fully utilizing its learned policy.

e. Best Epsilon

Among the given values, $\epsilon = 0.1$ provides the best performance because it maintains limited exploration while allowing the agent to exploit learned knowledge most of the time. The higher cumulative reward confirms its effectiveness in this scenario.

f. Conclusion

Therefore, $\epsilon = 0.1$ provides the best exploration–exploitation balance among the tested values. The results demonstrate that excessive exploration can reduce performance, while controlled exploration combined with exploitation can lead to higher cumulative rewards and better decision-making.

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results.

a. Introduction

Q-Learning and SARSA are important temporal-difference Reinforcement Learning algorithms used to learn the value of actions in different states. Both algorithms update their action-value function based on rewards and future values, but they differ in how the next action is considered during learning.

b. Observed Performance

In the given scenario, Q-Learning and SARSA are compared using their average rewards after **100, 200, and 300 episodes**. Q-Learning consistently achieves slightly higher average reward values than SARSA at all three stages. This indicates that Q-Learning is learning a more rewarding policy under the given conditions.

c. Q-Learning Behavior

Q-Learning is an **off-policy** algorithm. It updates its Q-value using the maximum estimated value of the next state, regardless of the action actually selected by the behavior policy. Its update equation is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

This allows Q-Learning to directly learn toward the optimal policy.

d. SARSA Behavior

SARSA is an **on-policy** algorithm. It updates its Q-value using the actual next action selected by the current policy. Therefore, its learning is influenced by the agent's exploration behavior. This can result in more conservative decisions, especially when exploration is significant.

e. Convergence Comparison

Since Q-Learning consistently produces slightly higher rewards at **100, 200, and 300 episodes**, it demonstrates better performance in the given experiment. The consistently higher reward suggests that Q-Learning is learning the optimal policy more effectively and may be **converging faster** toward a higher-performing policy.

However, reward values alone cannot mathematically prove faster convergence. A proper convergence comparison would require observing when the rewards become

stable and comparing the number of episodes required to reach a similar performance level.

f. Conclusion

Overall, **Q-Learning performs better than SARSA** in the given scenario because it consistently obtains higher average rewards. Its off-policy nature allows it to aggressively learn the optimal action, whereas SARSA considers the actual exploratory behavior. Thus, Q-Learning appears to converge faster under the given experimental conditions.

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance.

a. Introduction

The **discount factor (γ)** is an important parameter in Reinforcement Learning that determines how much importance an agent gives to future rewards compared with immediate rewards. Its value generally ranges from 0 to 1. A lower value makes the agent more focused on short-term outcomes, while a higher value encourages long-term decision-making.

b. Observed Data

In the given scenario, the RL agent is trained using discount factor values of **0.2, 0.5, and 0.9**. The average reward increases as the discount factor increases. This shows that considering future rewards improves the agent's overall performance in the given environment.

c. Effect of $\gamma = 0.2$

When $\gamma = 0.2$, the agent gives greater importance to immediate rewards and significantly discounts future rewards. This can be useful for problems where short-term decisions are more important, but it may prevent the agent from learning strategies that provide larger benefits in the long term.

d. Effect of $\gamma = 0.5$

With $\gamma = 0.5$, the agent considers both immediate and future rewards. It provides a moderate balance between short-term and long-term decision-making. However, the observed reward is still lower than that obtained with $\gamma = 0.9$.

e. Effect of $\gamma = 0.9$

With $\gamma = 0.9$, future rewards receive much greater importance. The agent therefore learns to select actions that may provide smaller immediate rewards but lead to better long-term outcomes. Since the average reward is highest at this value, $\gamma = 0.9$ is the most effective among the tested values.

f. Conclusion

The results show that increasing the discount factor improves long-term reward optimization in this scenario. Therefore, $\gamma = 0.9$ is the most suitable value among 0.2, 0.5, and 0.9 because it produces the highest average reward and encourages effective long-term decision-making.

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data.

a. Introduction

The **success rate** is an important performance metric in Reinforcement Learning because it indicates how often an agent successfully completes its assigned task. An increase in success rate during training generally shows that the agent is learning from its experiences and improving its policy.

b. Observed Data

In the given scenario, the success rate increases from **45% at 50 episodes to 92% at 200 episodes**. This represents a significant improvement of **47 percentage points**. The increase indicates that the agent has learned to select better actions and is becoming more effective at achieving its objective.

c. Learning Improvement

At 50 episodes, the success rate of 45% indicates that the agent is still learning and frequently makes unsuccessful decisions. After 200 episodes, the success rate reaches 92%, showing that repeated interaction with the environment has significantly improved its decision-making ability.

d. Policy Improvement

The increase in success rate suggests that the agent's **policy has improved**. The agent has learned to prefer actions that lead to successful outcomes and avoid actions that result in failure. This demonstrates effective learning through experience and reward feedback.

e. Optimal Policy Evaluation

A success rate of **92%** indicates a highly effective policy, but it does not by itself prove that the agent has learned a completely optimal policy. To confirm optimality, the success rate should remain consistently high over additional episodes and should ideally be compared with the theoretical maximum or performance of an optimal policy.

f. Conclusion

The increase from **45% to 92%** demonstrates substantial learning and policy improvement. The agent has likely learned a near-optimal policy because of its high success rate. However, additional testing and stable performance over more episodes are required to conclusively determine whether the policy is truly optimal.

Code of Conduct:

I KUSHITHA.G (192425329) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand